

16 Normalizing flows

Each layer has to be invertible!

$$16.1 \quad z \sim \text{Unif}[0,1] \Rightarrow P_r[z] = \begin{cases} 1, & 0 < z < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$x = f[z] = z^2 \Leftrightarrow z = f^{-1}[x] = \sqrt{x} \quad (x \in [0,1])$$

$$P_r[x] = \left| \frac{\partial f[z]}{\partial z} \right|^{-1} \cdot P_r[z] = \frac{1}{2z} \cdot P_r[z] = \frac{1}{2\sqrt{x}} \underbrace{P_r[\sqrt{x}]}_{=1 \text{ for } \sqrt{x} \in [0,1]} = \begin{cases} \frac{1}{2\sqrt{x}}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$16.2 \quad P_r[z] = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{z^2}{2}\right], \quad x = f[z] = \frac{1}{1 + \exp[-z]} \\ \Leftrightarrow x(1 + \exp[-z]) = 1 \\ \Leftrightarrow 1 + \exp[-z] = \frac{1}{x} \Leftrightarrow \log\left[\frac{1}{x} - 1\right] = z \\ \Leftrightarrow \log\left[\frac{x}{1-x}\right] = z$$

$$\left| \frac{\partial z}{\partial x} \right| = \left| \frac{1}{x(1-x)} \right| = \frac{\partial f^{-1}[x]}{\partial x}$$

$$P_r[x] = \left| \frac{\partial f[z]}{\partial z} \right|^{-1} \cdot P_r[z] \\ = \left| \frac{1}{x(1-x)} \right| \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{\log\left[\frac{x}{1-x}\right]^2}{2}\right]$$

$$16.3 \quad z = f^{-1}[x, \phi]$$

$$\frac{\partial f^{-1}[x, \phi]}{\partial x} = \frac{\partial f^{-1}[f^{-1}[\dots f^{-1}[x, \phi_k], \dots, \phi_2], \phi_1]}{\partial x} \\ = \frac{\partial f^{-1}[x, \phi_k]}{\partial x} \frac{\partial f^{-1}[\phi_k, \phi_{k-1}]}{\partial \phi_k} \dots \frac{\partial f^{-1}[\phi_2, \phi_1]}{\partial \phi_2}$$

$$\left| \frac{\partial f^{-1}[x, \phi]}{\partial x} \right| = \left| \frac{\partial f^{-1}[x, \phi_k]}{\partial x} \right| \dots \left| \frac{\partial f^{-1}[\phi_2, \phi_1]}{\partial \phi_2} \right| \\ = \left| \frac{\partial f[z, \phi]}{\partial z} \right|^{-1} \quad (\text{inverse of det. of forward mapping})$$

16.4

$$\Omega_1 = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}, \text{ diagonal, thus } \Omega_1^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & -\frac{1}{5} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{pmatrix}, \\ |\Omega_1| = 2 \cdot (-5) \cdot 1 \cdot 2 = -20$$

$$\Omega_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 \\ 1 & -1 & 2 & 0 \\ 4 & -2 & -2 & 1 \end{pmatrix}, \text{ triangular, thus } |\Omega_2| = 1 \cdot 4 \cdot 2 \cdot 1 = 8$$

$$\Omega_2^{-1}: \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & -1 & 2 & 0 & 0 & 0 & 1 & 0 \\ 4 & -2 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow{\substack{\text{II} - 2\text{I}, \\ \text{III} - \text{I}, \\ \text{IV} - 4\text{I}}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & -2 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 & -1 & 0 & 1 & 0 \\ 0 & -2 & -2 & 1 & -4 & 0 & 0 & 1 \end{array} \right)$$

$$\text{II} = \frac{1}{4} \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & -1 & 2 & 0 & -1 & 0 & 1 & 0 \\ 0 & -2 & -2 & 1 & -4 & 0 & 0 & 1 \end{array} \right) \xrightarrow{\substack{\text{II} \times 4, \\ \text{III} \times \frac{1}{2}}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 1 & 0 & -\frac{3}{4} & \frac{1}{8} & \frac{1}{2} & 0 \\ 0 & 0 & -2 & 1 & -5 & \frac{1}{2} & 0 & 1 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 1 & 0 & -\frac{3}{4} & \frac{1}{8} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 & -\frac{26}{4} & \frac{6}{8} & 1 & 1 \end{array} \right) \xrightarrow{\text{IV} + 2\text{III}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 1 & 0 & -\frac{3}{4} & \frac{1}{8} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 & -\frac{16}{4} & \frac{6}{8} & 1 & 1 \end{array} \right) \\ =: \Omega_2^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$

$$16.5 \quad \mathbb{E}[x] = \mathbb{E}[Az + b] = A \mathbb{E}[z] + \mathbb{E}[b] = A\mu + b$$

$$\text{cov}[x] = \text{cov}[Az + b] = A \text{cov}[z] A^T$$

$$16.6 \quad x = f[z] = Az + b, \quad P_r[z] = \text{Norm}_z[\mu, \Sigma]$$

$$z = A^{-1}(x - b) \Rightarrow \frac{\partial z}{\partial x} = A^{-1}$$

$$\Rightarrow P_r[x] = P_r[z] |A^{-1}|^{-1} \\ = \frac{1}{|\Sigma|^{1/2} \pi^{d_z/2}} \exp\left[-0.5(z - \mu)^T \Sigma^{-1}(z - \mu)\right] |A^{-1}|^{-1} \\ = \frac{1}{|A \Sigma A^T|^{1/2}} \exp\left[-0.5(z - \mu)^T A^{-1} \Sigma^{-1} A^{-1} A(z - \mu)\right] \\ = \frac{1}{|A \Sigma A^T|^{1/2}} \exp\left[-0.5(A(z - \mu))^T A^{-1} \Sigma^{-1} A^{-1} (A(z - \mu))\right] \\ = \frac{1}{|A \Sigma A^T|^{1/2}} \exp\left[-0.5(Az + b - A\mu - b)^T A^{-1} \Sigma^{-1} A^{-1} (Az + b - A\mu - b)\right] \\ = \frac{1}{|A \Sigma A^T|^{1/2} \pi^{d_x/2}} \exp\left[-0.5(x - A\mu - b)^T A^{-1} \Sigma^{-1} A^{-1} (x - A\mu - b)\right] \quad \square$$

$$16.7 \quad \text{ReLU}[u[z]] = \begin{cases} 0.1z & z < 0 \\ z & z \geq 0 \end{cases}$$

$$\text{ReLU}[u[x]] = \begin{cases} 10x & x < 0 \\ x & x \geq 0 \end{cases}$$

$$\frac{\partial \text{ReLU}[u[z_i]]}{\partial z_i} = \begin{cases} 0.1 & z_i < 0 \\ 1 & z_i \geq 0 \end{cases}$$

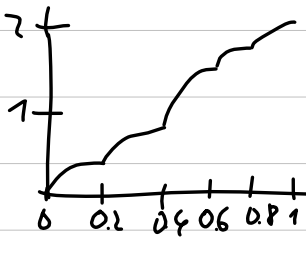
$$\Rightarrow \left| \frac{\partial \text{ReLU}[u[z]]}{\partial z} \right| = \prod_{i=1}^D \begin{cases} 0.1 & z_i < 0 \\ 1 & z_i \geq 0 \end{cases}$$

$$16.8 \quad \frac{\partial f[h_i]}{\partial h_i} = K \phi_i \Rightarrow \frac{\partial f[h]}{\partial h} = \text{diag}\left[\frac{\partial f[h_1]}{\partial h_1}, \dots, \frac{\partial f[h_D]}{\partial h_D}\right]$$

Again, the Jacobian is diagonal and the determinant just its product:

$$\left| \frac{\partial f[h]}{\partial h} \right| = \prod_{i=1}^D (K \phi_i) = K^D \prod_{i=1}^D \phi_i$$

16.9



noise is truncated

16.10

$$\frac{\partial h'}{\partial h} = \begin{pmatrix} \frac{\partial h'_1}{\partial h_1} & \frac{\partial h'_1}{\partial h_2} \\ \frac{\partial h'_2}{\partial h_1} & \frac{\partial h'_2}{\partial h_2} \end{pmatrix} \quad \frac{\partial h'_1}{\partial h_1} = 1 \quad \frac{\partial h'_1}{\partial h_2} = A \\ \frac{\partial h'_2}{\partial h_1} = B \quad \frac{\partial h'_2}{\partial h_2} = 1 + BA$$

$$(i) \text{ FV: } A, B \text{ dense: } \begin{pmatrix} 1 & A \\ B & 1 + BA \end{pmatrix}$$

$$(ii) \text{ elementwise flow: } A, B, \text{ diagonal (elementwise): } \begin{pmatrix} 1 & a & a \\ 0 & 1+a & a \\ 0 & 0 & 1+ba \end{pmatrix}$$

16.11

$$\hat{\phi} = \arg\min_{\phi} \left[\underbrace{\mathcal{L}\left[\frac{1}{N} \sum_{i=1}^N \mathcal{J}(x - f(z_i, \phi))\right]}_{= -\sum_{i=1}^N \log[q(f(z_i, \phi))]} \right]$$

$\rightarrow$ does not matter because it also up in the addition term that does not depend on ϕ

$\rightarrow$ as we generate the sample and also we know the latent variable